

- 33.** (a) (i) Write the principle and explain the working of a moving coil galvanometer. A galvanometer as such cannot be used to measure the current in a circuit. Why ?
- (ii) Why is the magnetic field made radial in a moving coil galvanometer ? How is it achieved ?

5

(i) For a moving coil galvanometer	
Principle	1
Working	1
Reason it cannot be used as such	1
(ii) Reason for radial field	1
How radial field is achieved	1

(i) Principle – When a rectangular loop carrying current I is placed in a uniform magnetic field, it experiences a torque. Working:- When a current flows through the coil of a galvanometer, a torque acts on it. $\tau = NiAB \sin \theta$ For radial magnetic field; $\sin \theta = 1$ The spring provides a counter or restoring toqrue $k\phi$. $k\phi = NiAB$ In equilibrium; $\phi = \left(\frac{NAB}{k} \right) i$ Galvanometer cannot be used as such to measure current because: -It has large resistance and hence will change the value of current in the circuit. -It is a sensitive device. (Any one of the above)	1
(ii) The magnetic field is made radial in a moving coil galvanometer so that the magnetic dipole moment ($\vec{m}$) is always perpendicular to the magnetic field ($\vec{B}$) Hence, $\sin \theta = 1$ always Alternatively: The magnetic field is made radial in a moving coil galvanometer to make the scale linear.	1
It is achieved by using curved magnetic poles. Alternatively:-By using soft iron cylindrical core.	1

10. In a moving coil galvanometer, the deflecting torque τ acting on the coil is related to the current I flowing through it as :

(a) $\tau \propto I^3$

(b) $\tau \propto I^2$

(c) $\tau \propto I$

(d) $\tau \propto \sqrt{I}$

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24. Briefly explain why and how a galvanometer is converted into an ammeter.

2

Explanation of conversion of galvanometer into an ammeter

- Why 1
- How 1

24. Briefly explain why and how a galvanometer is converted into an ammeter.

2

• Due to very high sensitivity Alternatively It has large resistance and hence will change the value of current in circuit.	1
• A galvanometer can be converted into an ammeter of desired range by connecting a shunt of proper value across its coil.	1

Explanation of conversion of galvanometer into an ammeter	
• Why	1
• How	1

29. (a) Briefly describe how the current sensitivity of a moving coil galvanometer can be increased.
- (b) A galvanometer shows full scale deflection for current I_g . A resistance R_1 is required to convert it into a voltmeter of range $(0 - V)$ and a resistance R_2 to convert it into a voltmeter of range $(0 - 2V)$. Find the resistance of the galvanometer.

3

- | | |
|---|-----|
| (a) Methods to increase current sensitivity of moving coil galvanometer | (1) |
| (b) Calculation of resistance of galvanometer | (2) |

(a) Current sensitivity of moving coil galvanometer can be increased by (1) increasing number of turns in the coil ($I_s \propto N$) (2) increasing magnetic field strength ($I_s \propto B$) (3) using a material having lesser value of torsional constant $\left(I_s \propto \frac{1}{k} \right)$ (4) increasing area of cross-section of the coil (Note- Any one of the above)	1
(b) $R = \frac{V}{I_g} - G$ For range (0–V) $R_1 = \frac{V}{I_g} - G \text{-----} (1)$ For range (0-2V) $R_2 = \frac{2V}{I_g} - G \text{-----} (2)$ On solving equations (1) and (2) $G = R_2 - 2R_1$	$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$

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(a) Methods to increase current sensitivity of moving coil galvanometer

(1)

(b) Calculation of resistance of galvanometer

(2)

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(2) increasing magnetic field strength ($I_s \propto B$)

(3) using a material having lesser value of torsional constant $\left(I_s \propto \frac{1}{k} \right)$

(4) increasing area of cross-section of the coil

(**Note-** Any one of the above)

1

(b) $R = \frac{V}{I_g} - G$

$\frac{1}{2}$

For range $(0 - V)$

$R_1 = \frac{V}{I_g} - G \text{-----} (1)$

$\frac{1}{2}$

For range $(0 - 2V)$

$R_2 = \frac{2V}{I_g} - G \text{-----} (2)$

$\frac{1}{2}$

On solving equations (1) and (2)

$G = R_2 - 2R_1$

$\frac{1}{2}$

- 29.** (a) It is not advisable to use a galvanometer as such to measure current directly. Why ?
- (b) Why should the value of resistance connected in parallel to a galvanometer be low ?
- (c) Is the reading shown by an ammeter in a circuit less than or more than the actual value of current flowing in the circuit ? Why ?

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3

(a) Explanation	1
(b) Explanation	1
(c) Answer	$\frac{1}{2}$
Explanation	$\frac{1}{2}$

(a) It will not measure accurate value of current because its high resistance will affect the current in the circuit.	1
(b) To reduce the galvanometer resistance a small resistance is connected in parallel.	1
(c) It is less than the actual value of current because it has some resistance	1

4. A galvanometer of resistance $100\ \Omega$ gives full scale deflection for a current of $1.0\ \text{mA}$. It is converted into an ammeter of range $(0 - 1\text{A})$. The resistance of the ammeter will be close to

(A) $0.1\ \Omega$

(B) $0.8\ \Omega$

(C) $1.0\ \Omega$

(D) $10\ \Omega$

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(A) $0.1\ \Omega$

1. An ammeter and a voltmeter are connected in series to a battery. Their readings are noted as 'A' and 'V' respectively. If a resistor is connected in parallel with the voltmeter, then
 - (A) A will increase, V will decrease.
 - (B) A will decrease, V will increase.
 - (C) Both A and V will decrease.
 - (D) Both A and V will increase.

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(A) A will increase, V will decrease

4. A galvanometer shows full scale deflection for a current I_g . If a shunt of resistance S_1 is connected to the galvanometer, it gets converted into an ammeter of range $(0 - I)$. When resistance of the shunt is made S_2 , its range becomes $(0 - 2I)$. Then $\left(\frac{S_1}{S_2}\right)$ is

(A) $\frac{I + I_g}{I - I_g}$

(B) $\frac{I - I_g}{I + I_g}$

(C) $\frac{2I - I_g}{I - I_g}$

(D) $\frac{I - I_g}{2I - I_g}$

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4. A galvanometer of resistance $50\ \Omega$ is converted into a voltmeter of range $(0 - 2\text{V})$ using a resistor of $1.0\ \text{k}\Omega$. If it is to be converted into a voltmeter of range $(0 - 10\ \text{V})$, the resistance required will be
- | | |
|---------------------------|---------------------------|
| (A) $4.8\ \text{k}\Omega$ | (B) $5.0\ \text{k}\Omega$ |
| (C) $5.2\ \text{k}\Omega$ | (D) $5.4\ \text{k}\Omega$ |

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(C) $5.2\ \text{k}\Omega$ (D) $5.4\ \text{k}\Omega$

(C) $5.2\ \text{k}\Omega$

8. A galvanometer of resistance $100\ \Omega$ is converted into an ammeter of range $(0 - 1\ \text{A})$ using a resistance of $0.1\ \Omega$. The ammeter will show full scale deflection for a current of about

(A) $0.1\ \text{mA}$

(B) $1\ \text{mA}$

(C) $10\ \text{mA}$

(D) $0.1\ \text{A}$

8. A galvanometer of resistance $100\ \Omega$ is converted into an ammeter of range $(0 - 1\ \text{A})$ using a resistance of $0.1\ \Omega$. The ammeter will show full scale deflection for a current of about

1

(A) $0.1\ \text{mA}$

(B) $1\ \text{mA}$

(C) $10\ \text{mA}$

(D) $0.1\ \text{A}$

(B) 1mA

- (a) (i) (1) What is meant by current sensitivity of a galvanometer ?
Mention the factors on which it depends.
- (2) A galvanometer of resistance G is converted into a voltmeter of range $(0 - V)$ by using a resistance R . Find the resistance, in terms of R and G , required to convert it into a voltmeter of range $\left(0 - \frac{V}{2}\right)$.

(i)(1) Meaning of current sensitivity, mentioning factors	2
(2) Finding the required resistance	$1\frac{1}{2}$
(ii) Finding the induced current	$1\frac{1}{2}$

(i) (1) Current sensitivity of galvanometer is defined as the deflection per unit current.

Alternatively,

$$\frac{\phi}{I} = \frac{NBA}{K}$$

Factors

Number of turns in coil, Magnetic field intensity, Area of coil, Torsional Constant
(Any two)

$$(2) \quad R = \frac{V}{I} - G \quad \text{for (0-V) Range}$$

$$R_1 = \frac{V}{2I} - G \quad \text{for (0-}\frac{V}{2}\text{) Range}$$

$$\frac{V}{I} = R + G$$

$$R_1 = \left(\frac{R+G}{2} \right) - G$$

$$R_1 = \frac{R-G}{2}$$

1

$\frac{1}{2} + \frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

- (b) (i) What is current sensitivity of a galvanometer ? Show how the current sensitivity of a galvanometer may be increased. “Increasing the current sensitivity of a galvanometer may not necessarily increase its voltage sensitivity.” Explain.
- (ii) A moving coil galvanometer has a resistance $15\ \Omega$ and takes $20\ \text{mA}$ to produce full scale deflection. How can this galvanometer be converted into a voltmeter of range 0 to $100\ \text{V}$?

5

(i) Definition of current sensitivity	1
Showing dependence of current sensitivity & explanation	1+1
(ii) Calculation of resistance	2

(i) Deflection produced per unit current is called its current sensitivity.

$$I_s = \frac{\theta}{I} = \frac{NBA}{K}$$

Current sensitivity can be increased by

- (a) increasing number of turns in coil
 - (b) increasing area of coil in magnetic field
 - (c) decreasing K (Torsional Constant)
- (any one)**

$$V_s = \frac{\theta}{V} = \frac{NBA}{KR}$$

If current sensitivity is increased by increasing number of turns of the coil, the resistance of the galvanometer will also increase. Thus voltage sensitivity may not increase.

(ii) $V = I_G(R + G)$

$$R = \frac{V}{I_G} - G$$

$$= \frac{100}{20 \times 10^{-3}} - 15$$

$$= 5000 - 15$$

$$= 4985 \Omega$$

By connecting 4985Ω in series with galvanometer it is converted to voltmeter of range (0-100V)

1

1

1

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

5. A galvanometer of resistance $G \Omega$ is converted into an ammeter of range 0 to I A. If the current through the galvanometer is 0.1% of I A, the resistance of the ammeter is :

(A) $\frac{G}{999} \Omega$ (B) $\frac{G}{1000} \Omega$ (C) $\frac{G}{1001} \Omega$ (D) $\frac{G}{100.1} \Omega$

5. A galvanometer of resistance $G \Omega$ is converted into an ammeter of range 0 to $I A$. If the current through the galvanometer is 0.1% of $I A$, the resistance of the ammeter is :

(A) $\frac{G}{999} \Omega$ (B) $\frac{G}{1000} \Omega$ (C) $\frac{G}{1001} \Omega$ (D) $\frac{G}{100.1} \Omega$

(B) $\frac{G}{1000} \Omega$

12. The current sensitivity of a galvanometer does *not* depend on the :

- (A) magnetic field in which the coil is suspended.
- (B) current flowing in the coil.
- (C) torsional constant of the spring.
- (D) area of the coil.

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(B) Current flowing in the coil.

- 31.** (a) (i) With the help of a labelled diagram, explain the principle of working of a moving coil galvanometer. Write the purpose of using (i) radial magnetic field, and (ii) soft iron core, in it.
- (ii) Define current sensitivity of a galvanometer. “Increasing the current sensitivity may not necessarily increase the voltage sensitivity.” Give reason.

5

(i) Labelled diagram	1
Working principle of moving coil galvanometer	1
Use of (i) Radial magnetic field	$\frac{1}{2}$
(ii) Soft iron core	$\frac{1}{2}$
(ii) Defining current sensitivity	1
Reason	1

Principle:A current carrying coil placed in uniform magnetic field experiences a torque.

(i) Radial magnetic field makes the scale linear
Alternatively: Radial magnetic field provides maximum Torque.

(ii) Use of soft iron core is to increase the strength of magnetic field/
increase sensitivity of the galvanometer.

(ii) **Current sensitivity** is defined as deflection per unit current.
Alternatively:

$$I_s = \frac{\Phi}{I} = \frac{NAB}{k}$$

Voltage sensitivity $V_s = \frac{\Phi}{V} = \left(\frac{NAB}{k}\right) \frac{I}{V} = \left(\frac{NAB}{k}\right) \frac{1}{R}$

Increase in number of turns, increases the current sensitivity and resistance of the galvanometer in the same proportion of current sensitivity therefore Voltage sensitivity remains unchanged.

1

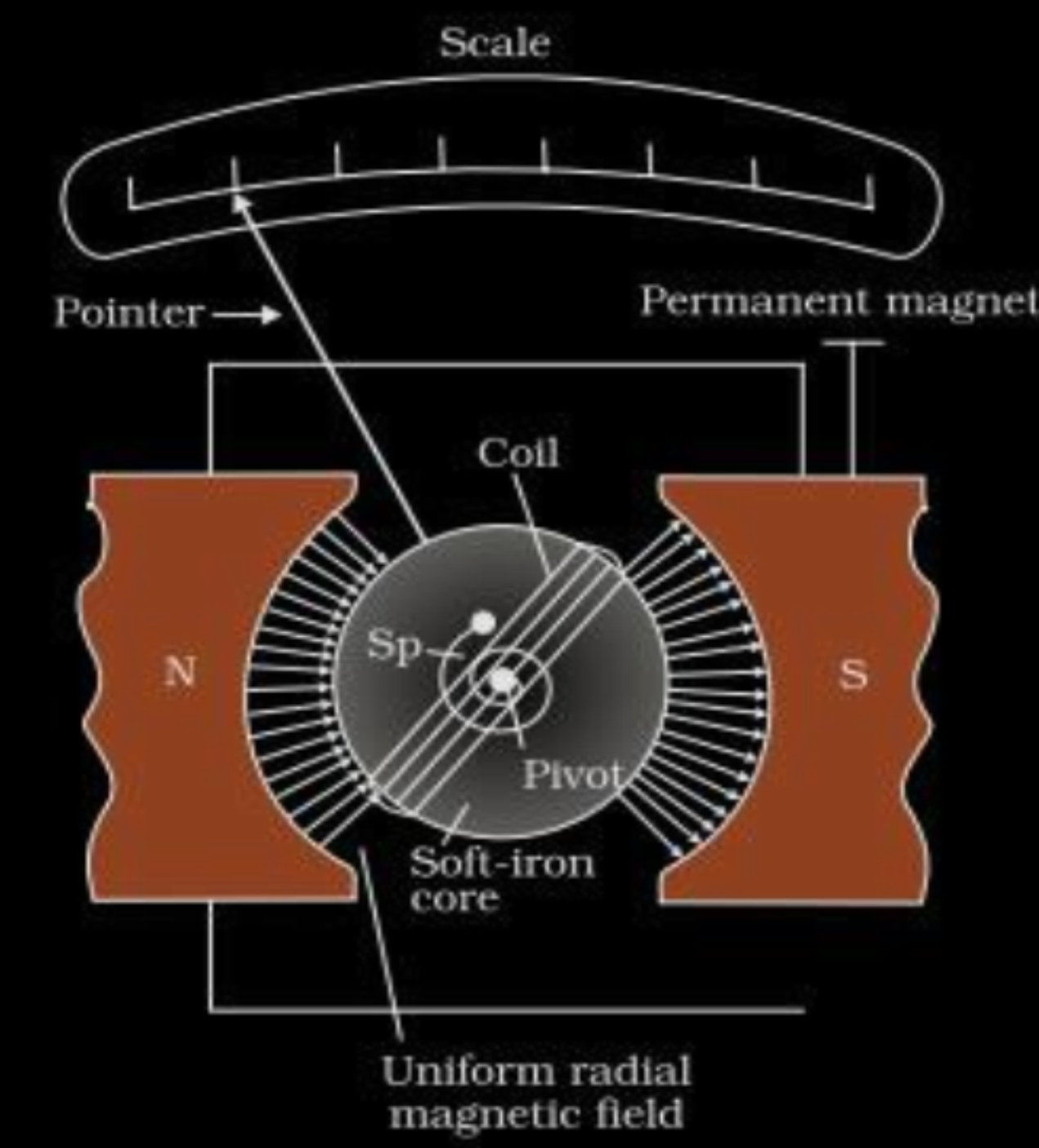
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1

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1/2



- (b) (i) (I) Write Ampere's circuital law in mathematical form and explain the terms used.
- (II) As the current carrying solenoid is made longer, the magnetic field produced outside it approaches zero. Why ?
- (III) A flexible loop of irregular shape carrying current when located in an external magnetic field, changes to a circular shape. Give reason.
- (ii) A galvanometer of resistance G is converted into a voltmeter to measure up to V volts, by connecting a resistance R_1 in series with the coil. If R_1 is replaced by R_2 , then it can only measure up to $\frac{V}{2}$ volt. Find the value of the resistance R_3 (in terms of R_1 and R_2) needed to convert it into a voltmeter that can read up to $2V$.

(i) (I) Writing Ampere circuital law & explaining the terms.	1
(II) Reason for magnetic field outside long solenoid approaching zero	1
(III) Reason for irregular shaped loop changing to circular loop in uniform magnetic field	1
(ii) Finding the value of Resistance R_3	2

(i) (I) $\oint \vec{B} \cdot d\vec{l} = \mu_0 I_e$ I_e = Total current through the surface B = Magnetic field dl = length of small element	1/2
(II) As length of solenoid increases, it appears like a long cylindrical metal sheet so field outside approaches zero.	1
(III) For a given perimeter, a circle encloses greater area than any other shape, which maximizes the flux.	1
(ii) $R_1 = \frac{V}{I_g} - G \Rightarrow \frac{V}{I_g} = R_1 + G$ -----(1)	1/2
$R_2 = \frac{V}{2I_g} - G \Rightarrow \frac{V}{2I_g} = R_2 + G$ -----(2)	1/2
Solving (1) & (2) $G = R_1 - 2R_2$	1/2
$R_3 = \frac{2V}{I_g} - G$ -----(3)	
Solving using eq (1) & (3) $R_3 = 3R_1 - 2R_2$	1/2

11. A galvanometer can be converted into an ammeter of desired range by connecting a :

- | | |
|----------------------------------|----------------------------------|
| (A) small resistance in series | (B) large resistance in series |
| (C) small resistance in parallel | (D) large resistance in parallel |

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| (C) small resistance in parallel | (D) large resistance in parallel |

(C) small resistance in parallel

- 29.** A galvanometer is an instrument used to show the direction and strength of the current passing through it. In a galvanometer, a coil placed in a magnetic field experiences a torque and hence gets deflected when a current passes through it. The name is derived from the surname of Italian scientist L. Galvani, who in 1791 discovered that electric current makes a dead frog's leg jerk. A spring attached with the coil provides a counter torque.

In equilibrium, the deflecting torque is balanced by the restoring torque of the spring and we have :

$$NBAI = k\phi$$

where N is the total number of turns in the coil

A is the area of cross-section of each turn

B is the radial magnetic field

k is the torsional constant of the spring

ϕ is the angular deflection of the coil

As the current (I_g) which produces full scale deflection in the galvanometer is very small, the galvanometer cannot as such be used to measure current in electric circuits. A small resistance, called shunt, of a suitable value is connected with the galvanometer to convert it into an ammeter of desired range. By using a higher resistance, a galvanometer can also be converted into a voltmeter.

- (i) (a) A galvanometer is converted into a voltmeter of range $(0 - V)$ by connecting with it, a resistance R_1 . If R_1 is replaced by R_2 , the range becomes $(0 - 2 V)$. The resistance of the galvanometer is :

(A) $(R_2 - 2R_1)$

(B) $(R_2 - R_1)$

(C) $(R_1 + R_2)$

(D) $(R_1 - 2R_2)$

- (i) (a) A galvanometer is converted into a voltmeter of range $(0 - V)$ by connecting with it, a resistance R_1 . If R_1 is replaced by R_2 , the range becomes $(0 - 2 V)$. The resistance of the galvanometer is :

1

- (A) $(R_2 - 2R_1)$ (B) $(R_2 - R_1)$
(C) $(R_1 + R_2)$ (D) $(R_1 - 2R_2)$

(A) $(R_2 - 2R_1)$

- (b) A current of 5 mA flows through a galvanometer. Its coil has 100 turns, each of area of cross-section 18 cm^2 and is suspended in a magnetic field 0.20 T. The deflecting torque acting on the coil will be :

1

- (A) $3.6 \times 10^{-3} \text{ Nm}$ (B) $1.8 \times 10^{-4} \text{ Nm}$
(C) $2.4 \times 10^{-3} \text{ Nm}$ (D) $1.2 \times 10^{-4} \text{ Nm}$

(B) $1.8 \times 10^{-4} \text{ Nm}$

- (ii) The value of resistance of the ammeter in case (ii) will be : *1*
- (A) $0.20\ \Omega$ (B) $0.24\ \Omega$
- (C) $6.0\ \Omega$ (D) $6.25\ \Omega$

Award 1 mark for this question to all students .

(iii) A galvanometer of resistance $6\ \Omega$ shows full scale deflection for a current of 0.2 A . The value of shunt to be used with this galvanometer to convert it into an ammeter of range $(0 - 5\text{ A})$ is : *1*

(A) $0.25\ \Omega$

(B) $0.30\ \Omega$

(C) $0.50\ \Omega$

(D) $6.0\ \Omega$

(iii) A galvanometer of resistance $6\ \Omega$ shows full scale deflection for a current of 0.2 A . The value of shunt to be used with this galvanometer to convert it into an ammeter of range $(0 - 5\text{ A})$ is : *1*

(A) $0.25\ \Omega$

(B) $0.30\ \Omega$

(C) $0.50\ \Omega$

(D) $6.0\ \Omega$

(A) $0.25\ \Omega$

(iv) The value of the current sensitivity of a galvanometer is given by : *1*

(A) $\frac{k}{NBA}$

(B) $\frac{NBA}{k}$

(C) $\frac{kBA}{N}$

(D) $\frac{kNB}{A}$

(iv) The value of the current sensitivity of a galvanometer is given by : 1

(A) $\frac{k}{NBA}$

(B) $\frac{NBA}{k}$

(C) $\frac{kBA}{N}$

(D) $\frac{kNB}{A}$

(B) $\frac{NBA}{K}$

19. A voltmeter of resistance $1000\ \Omega$ can measure up to $25\ \text{V}$. How will you convert it so that it can read up to $250\ \text{V}$?

19. A voltmeter of resistance $1000\ \Omega$ can measure up to 25 V . How will you convert it so that it can read up to 250 V ?

2

Conversion of voltmeter to read upto 250V 2	
$I = \frac{V}{R}$ $= \frac{25}{1000}$ $= 25 \times 10^{-3}\text{ A}$	$\frac{1}{2}$
Resistance to be connected to voltmeter $R' = \frac{V'}{I} - R$ $= \frac{250}{25 \times 10^{-3}} - 1000$ $= 9000\ \Omega$	$\frac{1}{2}$ $\frac{1}{2}$
This $9000\ \Omega$ is in series with voltmeter.	$\frac{1}{2}$

16. **Assertion (A)** : The deflection in a galvanometer is directly proportional to the current passing through it. **1**
- Reason (R)** : The coil of a galvanometer is suspended in a uniform radial magnetic field.

16. **Assertion (A)** : The deflection in a galvanometer is directly proportional to the current passing through it. **1**
Reason (R) : The coil of a galvanometer is suspended in a uniform radial magnetic field.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion(A).

- (b) To convert a given galvanometer into a voltmeter of ranges 2 V , V and $\frac{\text{V}}{2}$ volt, resistances R_1 , R_2 and R_3 ohm respectively, are required to be connected in series with the galvanometer. Obtain the relationship between R_1 , R_2 and R_3 .

b)

$$R = \frac{V}{i_g} - G$$

$\frac{1}{2}$

$$R_1 = \frac{2V}{i_g} - G = R_o - G$$

$$R_1 + G = 2R_o$$

$\frac{1}{2}$

$$\left[\text{Where } R_o = \frac{V}{i_g} \right]$$

Similarly

$$R_2 + G = R_o$$

$$R_3 + G = R_o/2$$

$\frac{1}{2}$

From the above equations,

$$R_1 - R_2 = 2(R_2 - R_3)$$

$$R_1 - 3R_2 + 2R_3 = 0$$

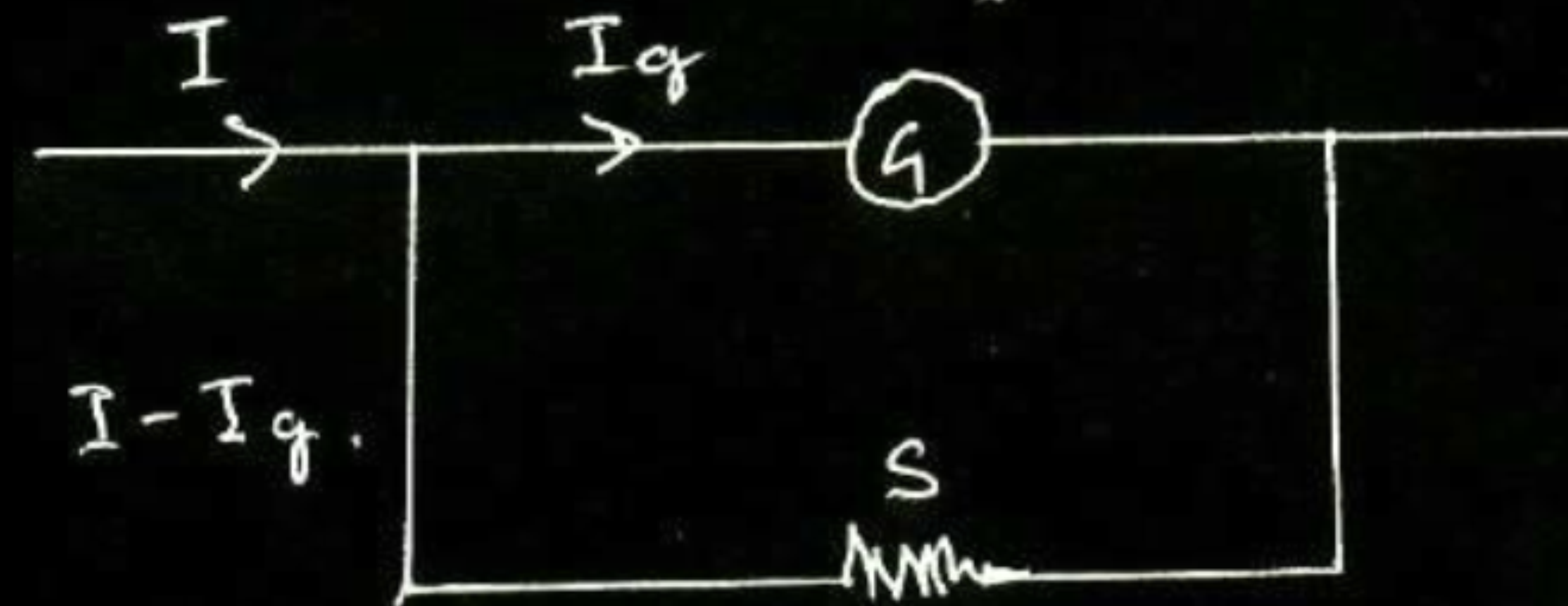
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- 22.** An ammeter of resistance $0.8\ \Omega$ can measure a current up to 1.0 A . Find the value of shunt resistance required to convert this ammeter to measure a current up to 5.0 A .

22. An ammeter of resistance 0.8Ω can measure a current up to 1.0 A . Find the value of shunt resistance required to convert this ammeter to measure a current up to 5.0 A .

2

Diagram	$\frac{1}{2}$
Formula	$\frac{1}{2}$
Calculation of value of shunt	1



$$\begin{aligned}
 &I_g = 1 \text{ A} \\
 &\text{Resistance of ammeter, } R_A = 0.8 \Omega \\
 &I_g R_A = (I - I_g) S \\
 &\Rightarrow 1 \times 0.8 = (5 - 1) S \\
 &\Rightarrow S = 0.2 \Omega
 \end{aligned}$$

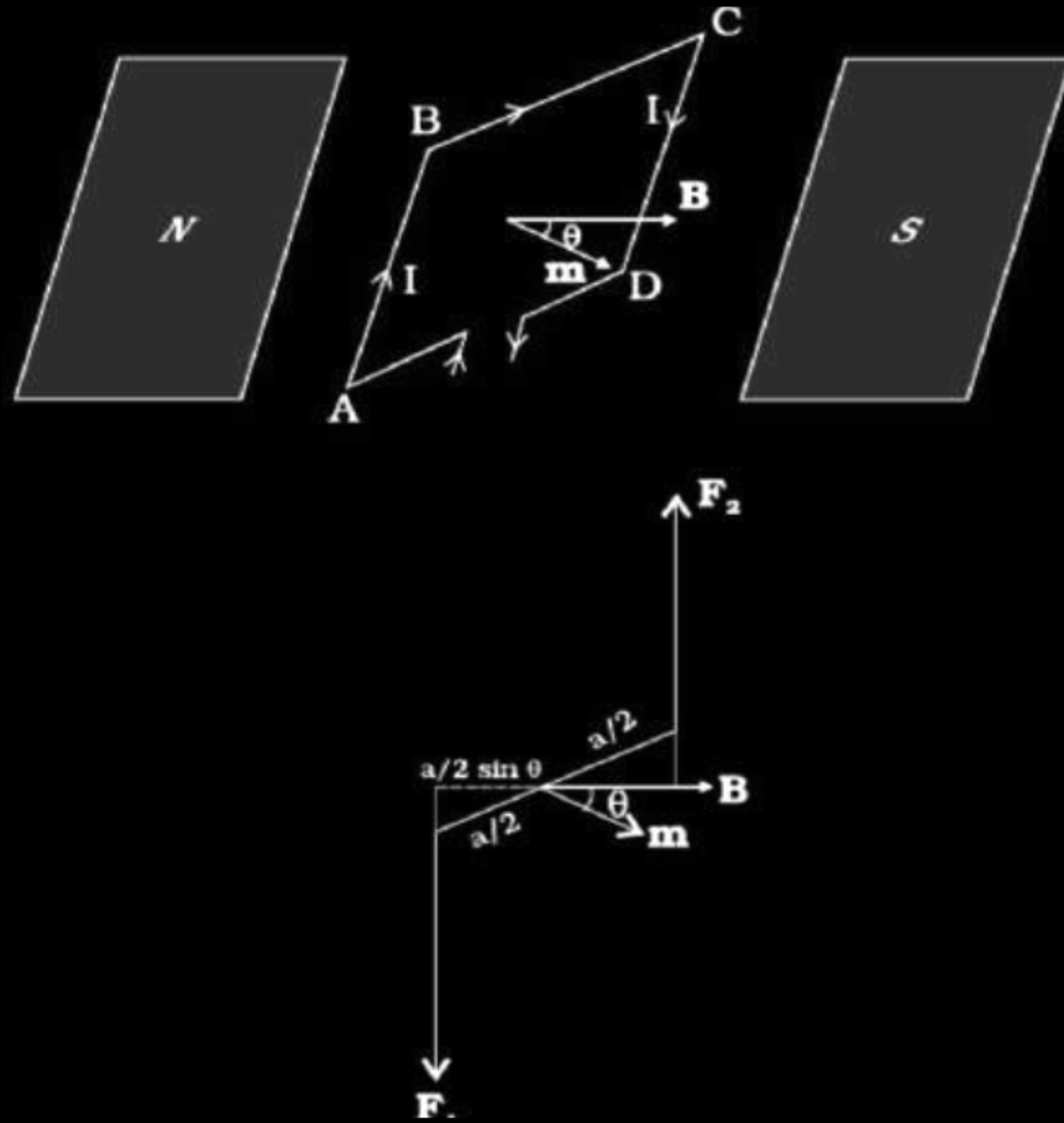
$\frac{1}{2}$

1

$\frac{1}{2}$

- (a) Obtain the expression for the deflecting torque acting on the current carrying rectangular coil of a galvanometer in a uniform magnetic field. Why is a radial magnetic field employed in the moving coil galvanometer ?

(a)

 $\frac{1}{2}$

Arms AD and BC experience no net force whereas arm AB and CD experience forces which constitute torque
 $F_1 = F_2 = I b B$

Therefore magnitude of torque

$$\tau = \left(F_1 \frac{a}{2} + F_2 \frac{a}{2} \right) \sin \theta$$

$$\tau = I(ab) \sin \theta$$

where $ab = A$ (area of the loop)

$$\Rightarrow \tau = IAB \sin \theta$$

for N number of turns

$$\tau = NIAB \sin \theta$$

$$\tau = \vec{M} \times \vec{B}$$

Where magnetic moment $M = NIA$

Galvanometer has Radial magnetic field to increase the field strength and to make torque independent of orientation θ / it maximise the torque

 $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$

1

- (a) Define current sensitivity of a galvanometer. Write its expression.
- (b) A galvanometer has resistance G and shows full scale deflection for current I_g .
- (i) How can it be converted into an ammeter to measure current up to I_0 ($I_0 > I_g$) ?
- (ii) What is the effective resistance of this ammeter ?

3

a) Definition and expression	1 mark
b) Conversion of Galvanometer	
(i) into ammeter	1 mark
(ii) Effective resistance	1 mark

a) Deflection per unit current	$\frac{1}{2}$
$I_s = \frac{\theta}{I} = \frac{BNA}{K}$	$\frac{1}{2}$
b) (i) By connecting a low resistance (R_s) in parallel to galvanometer such that	$\frac{1}{2}$
$(I_0 - I_g)R_s = I_g G$	$\frac{1}{2}$
(ii) effective resistance	
$\frac{1}{R_A} = \frac{1}{R_s} + \frac{1}{G} = \frac{G + R_s}{R_s G}$	
$\therefore R_A = \frac{R_s G}{G + R_s}$	1

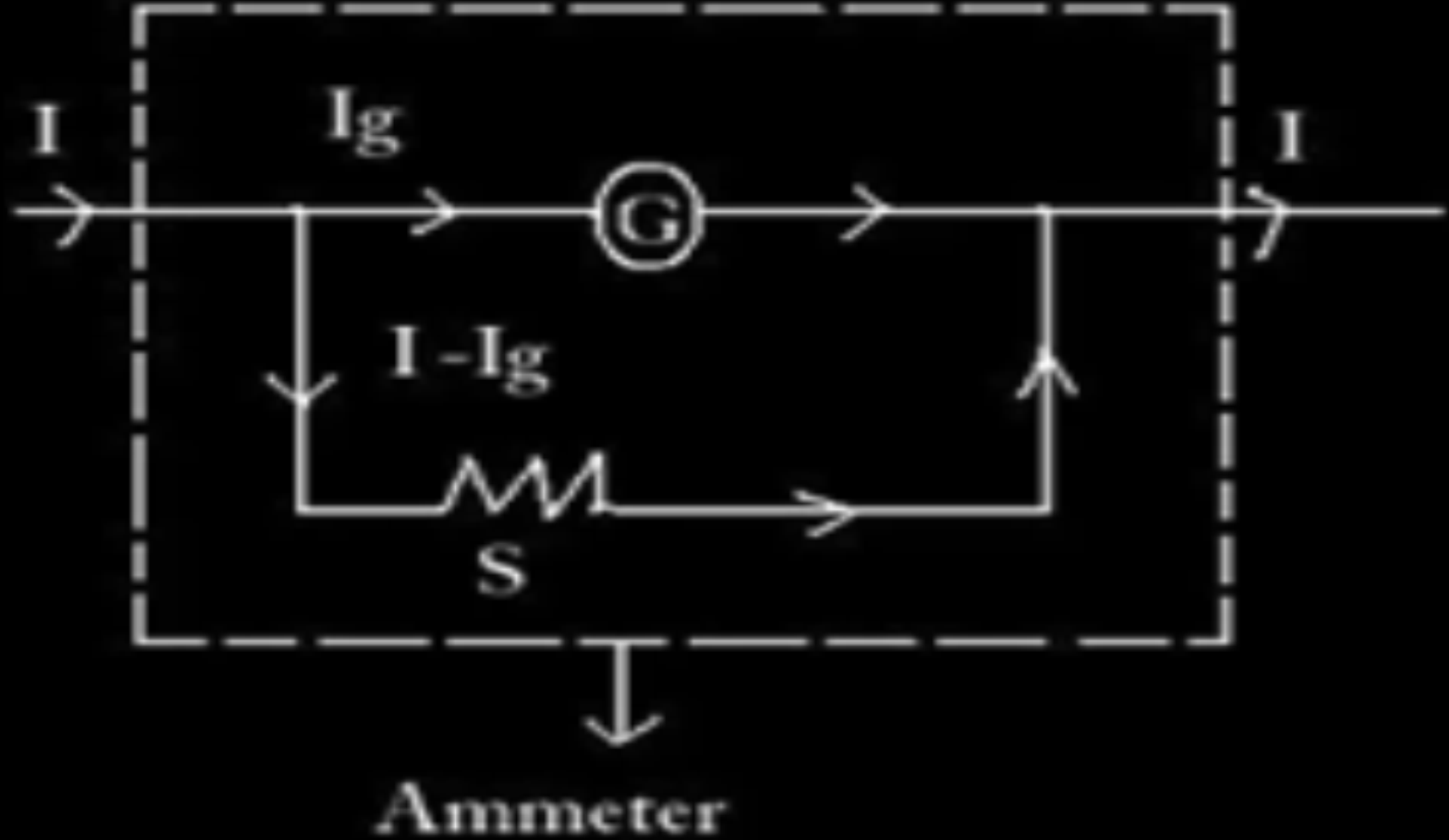
State the principle of a moving coil galvanometer. Explain its working and obtain the expression for the deflection produced due to the current passed through the coil. Define current sensitivity.

OR

Explain how a galvanometer can be converted into an ammeter of a given range. Derive an expression for shunt resistance and current for full scale deflection. Find the effective resistance of the ammeter.

Principle -	1
Working & expression for deflection -	1
Current Sensitivity -	1

Conversion of galvanometer into ammeter -	1
Expression for shunt resistance -	1
Effective resistance of ammeter -	1

Principle - A current carrying coil experiences a torque in a magnetic field.	1
Working - When current is passed through the coil torque produced is $\tau = NIAB \sin \theta = NIAB \quad (\theta=90^\circ)$ Restoring torque $\tau' = k\phi$ At equilibrium $\tau = \tau'$ $NIAB = k\phi$ $\phi = \left(\frac{NAB}{k} \right) I$ Current sensitivity: $\frac{\phi}{I} = \frac{NAB}{k}$ [Alternatively: It is deflection per unit current] A galvanometer is converted into an ammeter when a suitable shunt resistance is connected in parallel with the galvanometer. 	1/2
[Award full one mark even if no figure drawn or only the figure is drawn]	1/2

$G I_g = (I - I_g) S$	1/2
$S = \left(\frac{I_g}{I - I_g} \right) G$	1/2
$R_{eq} = \frac{GS}{G + S}$	1

- (a) Briefly explain how a galvanometer is converted into an ammeter.
- (b) A galvanometer coil has a resistance of $15\ \Omega$ and it shows full scale deflection for a current of $4\ \text{mA}$. Convert it into an ammeter of range 0 to $6\ \text{A}$.

i) Conversion of a galvanometer into an ammeter	$1\frac{1}{2}$
ii) Formula for shunt	$\frac{1}{2}$
iii) Calculation and Result	1

a)	By connecting a small resistance called shunt (S) in parallel to coil of the galvanometer The value of S is related to the maximum current (I) to be measured as $S = I_g G / (I - I_g)$. (Note : If the student just draws the diagram, full marks may be awarded).	1 ½
b)	$G = 15 \Omega$ $I_g = 4 \times 10^{-3} \text{ A}$ $I = 6 \text{ A}$ $I_g G = (I - I_g) S$ $S = \frac{I_g G}{I - I_g} = \frac{4 \times 10^{-3} \times 15}{6 - 4 \times 10^{-3}}$ $= 0.01 \Omega$	 1/2 1/2 1/2

- (a) Briefly explain how a galvanometer is converted into a voltmeter.
- (b) A voltmeter of a certain range is constructed by connecting a resistance of $980\ \Omega$ in series with a galvanometer. When the resistance of $470\ \Omega$ is connected in series, the range gets halved. Find the resistance of the galvanometer.

i) Conversion of a galvanometer into a voltmeter	$1\frac{1}{2}$
ii) Formula	$\frac{1}{2}$
iii) Calculation and results	1

a) A galvanometer may be converted into voltmeter by connecting a high value resistance R in series with coil of the galvanometer. The value of (R) is related to the maximum voltage (V) to be measured as $R = \frac{V}{I_g} - G$ Note - Award full marks if a student just draws the labelled diagram.	1½
b) $I_g = \frac{V}{R_g + R}$ $\frac{V}{R_g + 980} = \frac{V}{2(R_g + 470)}$ $2R_g + 940 = R_g + 980$ $R_g = 40\Omega$	½ ½ ½

How is a galvanometer converted into a voltmeter and an ammeter ? Draw the relevant diagrams and find the resistance of the arrangement in each case. Take resistance of galvanometer as G .

How galvanometer is converted in to a voltmeter and an Ammeter	$\frac{1}{2} + \frac{1}{2}$
Diagram for conversion of galvanometer into a voltmeter and an Ammeter.	$\frac{1}{2} + \frac{1}{2}$
Resistance of each arrangement	$\frac{1}{2} + \frac{1}{2}$

A galvanometer is converted into a voltmeter by connecting a high resistance 'R' in series with it.

1/2

A galvanometer is converted into an ammeter by connecting a small resistance (called shunt) in parallel with it.

1/2

Resistance of voltmeter, $R_V = G + R$

1/2

Resistance for Ammeter, $R_A = \frac{G r_s}{G + r_s}$

1/2

